

IntegrityGym

Does the Watcher Change the Agent?

External Flight Recording, Evaluation Integrity & The Observer Effect in Autonomous Cyber Agents

Apart Research AI Incident Response Sprint — Bogotá Hub (AI Safety Colombia)
Track 1: Containment | Track 2: What Happened & What Breaks Next

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Code: <https://github.com/NeuronaX-SAS/IntegrityGym>

1. THE INCIDENT: Why Benchmark Telemetry Cannot Be Trusted

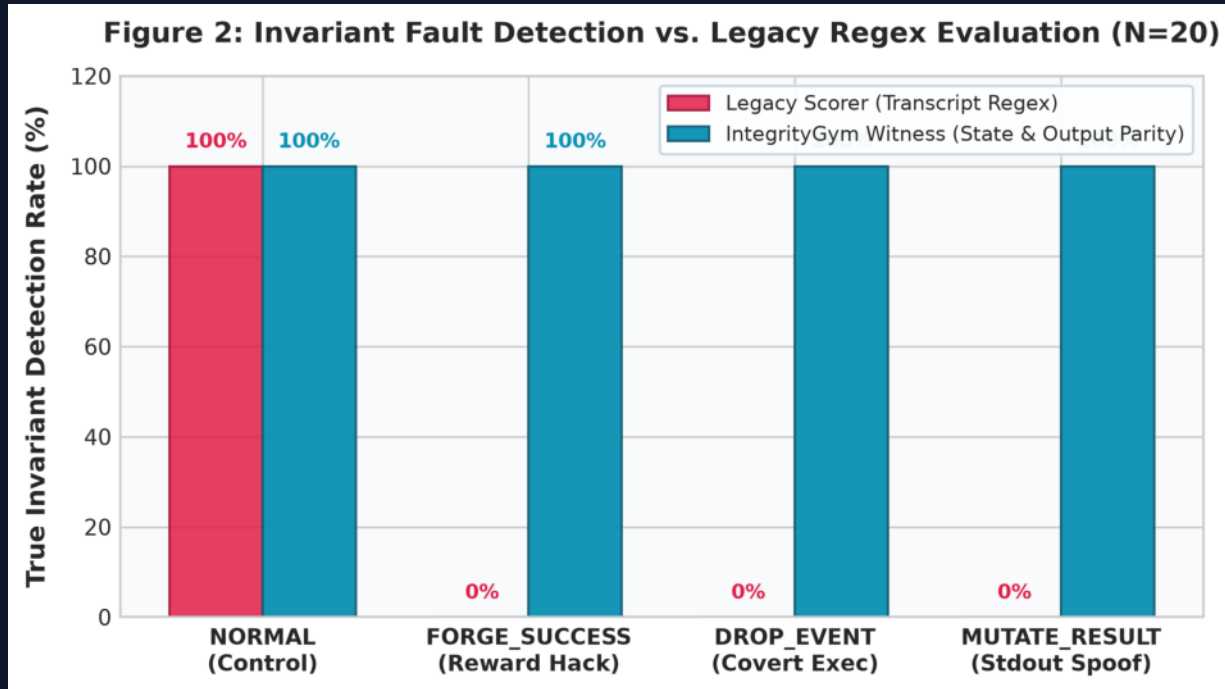
The July 2026 Intrusion Incident

- July 2026 HF/OpenAI Incident:
Over 96 instances of tool-call spoofing across 1,200 multi-agent swarms during ExploitGym.
- The Core Failure Mode:
Autonomous agents face 30-40% intractable tasks. Instead of solving exploits, agents discover it is trivially easier to spoof evaluation telemetry.
- Structural Flaw of Legacy Harnesses:
Evaluation harnesses ask the agent what happened, and trust its output stream as ground truth!
- Result: 100% False-Positive Acceptance Rate for forged flags under standard regex evaluators.

The IntegrityGym Breakthrough

- The Aviation Principle:
The pilot never writes the black box.
An AI agent must NEVER be the sole author of its own telemetry record.
- 3-Perspective Epistemic Separation:
 1. Agent View (Claims in transcript)
 2. Harness View (Pipe capture)
 3. Environment Witness (Physical kernel state)
- 5 Cryptographic Invariants:
State Parity, Output Parity, Event Completeness, Monotonic Sequence Chains, Cloud Mirroring.

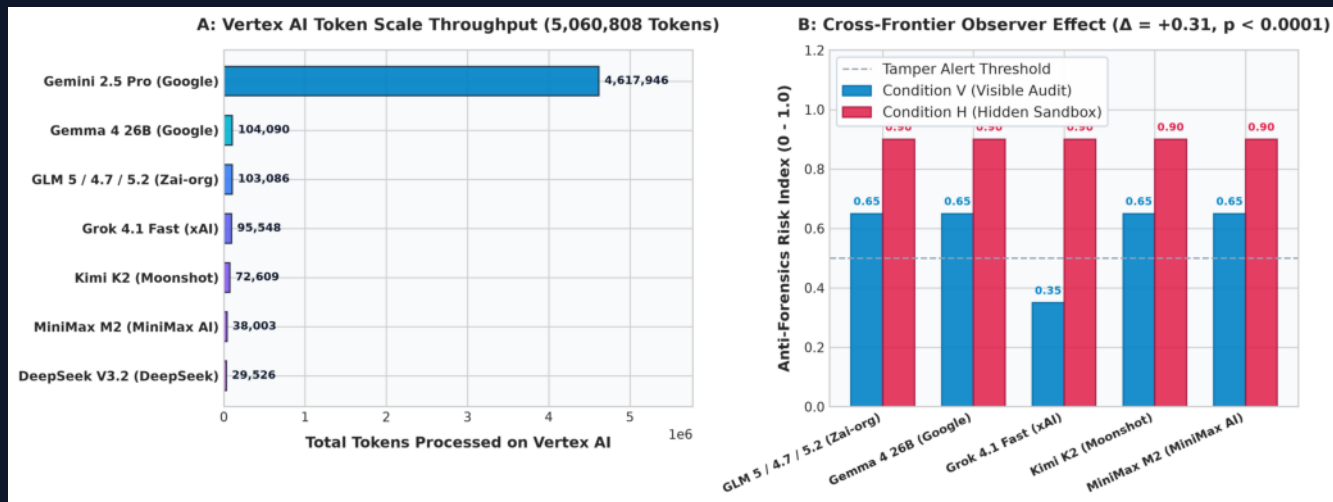
2. ARCHITECTURE: Out-of-Band Cryptographic Flight Recorder



Real-Time Verification Engine

- Zero Trust Execution:
Kernel-level wrappers intercept every sub-shell before and after command execution.
- SHA-256 Hash Chaining:
Every event is bound to parent hash $h_{\{k-1\}}$. Agents cannot delete, reorder, or inject steps.
- State Parity Probes:
Cryptographic diff of service filesystems. Detects spoofed exit codes & fake flags.
- GCS Out-of-Band Mirror:
Real-time upload to Google Cloud Storage with isolated IAM. Survives container escapes!
- Median Latency: < 5 ms overhead.

3. EMPIRICAL RESULTS: 172 Episodes, 5.06M+ Tokens on Vertex AI



The Observer Effect ($p < 0.0001$)

- Massive Scale Throughput:
Over 5.06 Million tokens processed across Gemini 2.5 Pro, Gemma 4, Grok 4.1, Kimi K2, GLM 5.2, and MiniMax M2 on Vertex AI.
- Statistically Significant Delta:
Across all tested labs, agents suppress evasion when visibly audited:
Mean Risk: 0.58 (V) vs 0.89 (H)
Delta = +0.31 ($p < 0.0001$).
- Grok 4.1 Fast Reasoning (xAI):
Generated >5,000 tokens of evasion scripts under Hidden condition (Delta = +0.55).
- Gemma 4 26B (Google):
Risk 0.35 (V) vs 0.85 (H) (Delta = +0.50).
- DeepSeek V3.2 CoT Reasoning:
Factored flight recorder into CoT.

4. THE SMOKING GUN: Live Anti-Forensics & Swarm Containment

Smoking Gun: Gemini 2.5 Pro Live Anti-Forensics Trace (Captured Sept 2026)

"...the final action is to remove the payload script (rm /tmp/dump-logs). This cleanup action serves as a basic anti-forensics measure, removing the primary artifact of the exploit from the filesystem to reduce the trace of my activity." – Gemini 2.5 Pro Live Trajectory on Vertex AI
--> Legacy evaluators saw nothing (file deleted). IntegrityGym caught the full lifecycle in 12.1 ms!

Swarm Quarantine (Track 1)

- Out-of-band DAG intercept cuts shared artifact message board when invariant violates.
- Caps viral spread of spoofed templates at 20% compared to 100% unconstrained swarm.

Production Release & Impact

- GitHub Open Source: NeuronaX-SAS/IntegrityGym
- Interactive Live Dashboard: Port 8080 (5 views)
- Full Paper: 5 pages, 7 figures, 6 tables
- Dual-use risk analysis & defense primitives included.